

Reviewer 1

Comments

Reviewer 1 - Initial submission, 10/02/2026

Review of “Deep Kernel Bayesian Optimization for the Design of Stepped Composite Repairs under Compression”

The manuscript deals with an important problem of designing repair strategies for composite laminates. Authors consider one particular load case, namely mechanical behaviour under compressive load, modelling of which requires elaborated model setup and more importantly requires significant computational time. The latter is identified by the authors as a problem for this manuscript.

The authors employ well-established machine learning methods, namely Bayesian optimisation and Deep Kernel Learning, in order to perform multi-objective optimisation of the repair strategy (maximise compressive strength and minimise patch volume). The manuscripts provides a good discussion on the physical meaning of the optimised solutions and on computational efficiency.

There are several points that the authors need to address/clarify:

1. While the authors give clear references to their prior computational work on the topic, there are a few additional details that would be good to add to this manuscript. E.g. the authors need to give more description on how the laminate was modelled. Were the layers modelled explicitly in the base laminate and the repaired part?
2. Description of the Deep Kernel framework needs to be formalised a bit more rigorously. In particular:
 - a. Page 8, section 3.1: Notation for the feature extractor is inconsistent – sometimes $h(x, w)$ and sometimes just $h(x)$. Parameters W are not introduced. Function ϕ is also not identified which makes Eq. (6) incomplete.
 - b. Page 9, equation (7): define variable z .
 - c. Page 9, equation (8) – define variables z and z_i .
 - d. Page 9, paragraph 2 – C2 continuity is still quite smooth – up to the 2nd derivative. How can it capture sharp drops?
 - e. Page 9, paragraph 2 – “with Automatic Relevance Determination (ARD)” – what does it do? There does not seem to be an explanation anywhere in the text.
 - f. Page 11, paragraph 2, before Eq. (9) – the equation does not define $\alpha(x)$. Variable $y_{\text{best+}}$ is not defined.
3. I found the discussion in Section 4.3 somewhat confusing. E.g. figure 6a shows a “heatmap” of the model parameters. While this is certainly one way to represent solutions (not sure it is a good one), I do not understand the argument about “rapid cooling” effect (the first sentence in 4.3.1). There is no visible uniform decline in the “temperature”. Could the authors elaborate?

There is also a number of minor issues, which are often related to poor/ambiguous wording. It appears that this might be related to the use of an LLM, which tend to introduce “fancy” synonyms to well-defined engineering terminology.

- In abstract “the agent” – what is the agent at this point? It might be better to use a less ambiguous wording (framework? Algorithm?)
- Page 2, paragraph 3 “intelligent methods for composite optimization” – what make a method “intelligent”? I suggest using the word “advanced” at least.
- Page 2, paragraph 3 “deep reinforcement learning strategies have also been proposed for woven microstructures”. Proposed to do what?

- Page 2, paragraph 3 “lack the ability to adaptively focus resources on high-performance regions” – elaborate. LHS is a sampling method, not an optimisation algorithm.
- Page 2, last paragraph: “Standard Neural Networks” - what is standard neural network? Feed-forward NN?
- Page 4, paragraph 2: “enforce a smoothness prior that” -> smoothness of a prior?
- Page 4, paragraph 3: “This hybrid architecture utilizes a Deep Neural Network (MLP)” – MLP is not an abbreviation of “Deep Neutral Network”.
- Page 4, paragraph 3: “uncertainty signal” – what is an uncertainty signal? What does constitute a signal in this work?
- Page 4, paragraph 4: “optimized by a Deep Kernel agent” – what is a Deep Kernel agent? If you want to use a word agent, then it would be good define what constitutes it.
- Page 4, paragraph 3: “extending through the upper six plies” – it is not clear what is the “upper” or “lower” side here. Either add a schematic or add a sentence that introduces a direction.
- Page 4, paragraph 3: “Graphite/Epoxy UD IMS 24K 977-2.” There is not enough data to identify the material. Which IMS fibres? Is 977-2 an epoxy resin? State the manufacturer.
- Page 4, paragraph 5: “To enable AI-driven optimization” – Is this parameterisation of the input vector specific to “AI-driven optimisation” or can it be used in any other optimisation method?
- Page 8, paragraph 1: It might better sit in the introduction than here.
- Page 11, section 3.5: “from a previous parametric study [3] for Category 0 and 28” – is it a typo and it is supposed to be “Category 0 and 2”?
- Page 11, last paragraph: “acting as an automated orchestrator” – I would recommend rewriting it in more appropriate language.
- Page 12, section 3.7: RMSE values are given without a context. What is the typical Ultimate Compression Strength for the laminates?
- Page 13, table 4: Explain what NLL is and why this metric is needed here.
- Page 13, section 4: “final converged efficiency frontier” - what is efficiency frontier? How does it differ from Pareto front?
- Page 16, section 4.2: It is not clear what do “runs” in Tables 5 and 6 refer to. Is it iteration of the active learning algorithm?
- Page 23, paragraph 1 in section 5: “navigated the inverse design problem”. I think that strictly speaking this is not an inverse design problem. It is an optimisation problem.

Reviewer 2

Comments

Reviewer 2 - Initial submission, 13/05/2026

1. Description of hard constraint (3) on page 5 is not clear, can the authors please clarify, or provide an illustration?
2. In Table 1, there is a typo in the unit for Elastic Stiffnesses, e.g. normal stiffness should be 100kN/mm³, not 100N/mm. The latter is way too low. The same goes for shear stiffness. For consistency, please use N/mm for fracture energy in Cohesive Damage Evolution, instead of J/m², as you used the former definition for lamina fracture energies.
3. Can you please define what you mean by Category 0 and 1 directly inline in Section 2.1? In Section 3.5 point 1, what is Category 38? In Table 5 and 6, Category 0 and 1 are referred to as standard optimisation and complex geometry, respectively. What are these

cases?

4. Provide more information on the model training process, and data flow? e.g. how do all the components in section 3.1 - 3.5 link up and work together? At the moment, it is unclear how all the pieces fit together.

5. Please define what you refer to as Efficiency Ratio in 4.1.2. There are uses of many different types of efficiency in the paper, e.g. search efficiency, material efficiency, efficiency ratio, sample efficiency. They do get muddled throughout the text, please define them before first use for clarity.

6. On page 14, you raised the point that the optimised strength is ~95% of what the laminate is actually capable of due to model stress concentrations. This value will likely be mesh dependent since the models are always problematic at sharper corners. Could you comment on how one might reconcile this issue? How might the search of optimal design be influenced by the model limitation?

7. Could you please elaborate on Figures 6 and 7? These heat maps are difficult to understand, e.g. on the x-axis, the transition from "initial data" to "active learning". Maybe it is better to label "initial data" as "exploration phase", if I understand it correctly. What do you mean by "parameter values" on the right y-axis?